

Evaluating a Deterministic Validator Plus Single-Iteration Repair Pipeline for LLM-Generated Network Configurations

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Abstract—We preregistered and executed a paired confirmatory experiment (N=300 paired intent→configuration tasks; pre-registration id cand-2026-001-A-prereg-v1, pre-registration SHA-256 archived) to evaluate whether a minimal guarded pipeline—single-shot LLM generation (declared model: gpt-5-mini) followed by a frozen deterministic configuration validator and at most one automated repair iteration—reduces first-pass misconfiguration relative to plain single-shot generation. Execution completed 600 episodes and used 301 model calls. Observed outcomes: baseline = 299/300 valid (99.667%); guarded = 300/300 valid (100.0%); paired counts n11=299, n10=1, n01=0, n00=0. Exact McNemar two-sided $p = 1.00$; absolute paired difference = 0.0033333; paired bootstrap 95% CI = [0.00, 0.01] (10,000 resamples, seed 1729). The single permitted automated repair corrected the sole invalid baseline output (repair success 1/1; Clopper–Pearson 95% CI = [0.025, 1.0]). The preregistered planning assumption used for power calculations (planned baseline pass rate = 0.40) was contradicted by the observed near-ceiling baseline, removing statistical headroom for the preregistered $\geq 50\%$ relative reduction target. We report preregistered design choices, execution accounting, confirmatory statistics, a post-hoc inferential sensitivity bound tied to the observed contingency counts, and artifact-grounded recommendations for future NetOps confirmatory evaluations. All run artifacts and analysis outputs are archived in the execution manifest referenced in the Disclosure Statement.

I. INTRODUCTION

Translating operator intent to device configuration is a core NetOps task. Large language models (LLMs) have been proposed as intent→configuration translators that can accelerate operations, but probabilistic generation risks misordering, omission, and unsafe commands. Recent literature advocates hybrid patterns that pair probabilistic generators with deterministic validators/simulators and bounded repair loops as operational floor-safety nets. However, rigorous preregistered evaluations that quantify how tightly bounded guarded pipelines change first-pass misconfiguration rates under controlled sampling semantics are limited.

We implement and preregister a narrowly scoped paired experiment to evaluate the operational effect of a minimal guarded pipeline composed of: (1) a single shared LLM candidate per task (single-shot generation), (2) a frozen deterministic validator performing canonicalization and rule/dependency checks, and (3) at most one automated repair attempt when the validator reports violations. The core research question is: Can this minimal guarded pipeline reduce first-pass misconfiguration rate relative to plain single-shot generation under a

controlled paired design? Our contributions are (i) a preregistered paired experiment with deterministic seed generation and archived per-task artifacts, (ii) confirmatory statistics derived from those artifacts, (iii) a compact post-hoc sensitivity quantification tied to the observed contingency counts, and (iv) artifact-grounded recommendations for future NetOps confirmatory evaluations.

II. RELATED WORK

Recent work on LLMs for NetOps and intent-driven configuration has produced numerous system demonstrations and surveys documenting both promise and concrete failure modes. Empirical demonstrations of intent→configuration translation report that LLMs can synthesize plausible device configuration templates in lab and testbed settings, yet they frequently surface operationally relevant errors such as command misordering, omission of required safety steps, and overconfident assertions that lack mechanical verification (example system evaluation and discussion: Nguyen et al. 2024 [10.1002/nem.2313]). Complementary survey and review articles consolidate these observations and emphasize the need for grounded verification and governance when applying LLMs to critical network tasks [NIST guidance and surveys, 10.6028/nist.ai.100-2e2023].

Parallel technical threads motivate guarded architectures. Work in adjacent domains (clinical QA, code repair, and general RAG/verification research) shows measurable reductions in factual errors when probabilistic generators are paired with retrieval, deterministic checks, or repair loops; these findings have been ported by several NetOps proposals arguing for generate→verify→repair pipelines rather than blind single-shot deployment [e.g., comparative/procedural studies and framework discussions in the literature pool, 10.36227/techrxiv.176784505.56675782/v1; 10.2139/ssrn.6608518]. Architectural proposals for intent-aware automated remediation and multi-agent NetOps orchestration similarly argue that deterministic validators and bounded repair/backout policies are central to safe agentic NetOps workflows (see recent agentic NetOps proposals and intent-translation demonstrations, e.g., 10.36227/techrxiv.177004277.71795055/v1).

Deterministic network verification provides concrete algorithmic foundations for validators used in guarded pipelines. Prior work on specification-based and header-space analyses demonstrates tractable static checks for reachability, ac-

cess control, and update-safety properties without requiring live device execution; such algorithms can be integrated to canonicalize generated artifacts into normalized ASTs and to flag violations that trigger repair logic [Beckett et al., 2017 foundation for incremental/specification checks, 10.1145/3098822.3098834]. These deterministic checks differ fundamentally from empirical or emulation tests: they provide provable, rule-based pass/fail decisions for a defined specification, but they do not by themselves measure runtime control-plane or packet-level effects unless combined with higher-fidelity simulation or emulation.

Limitations in the prior literature motivate the present work. Many demonstrations and proposals are system-level or exploratory rather than preregistered confirmatory experiments; reported improvements from RAG, verifier-assisted generation, or agentic architectures often lack paired within-task statistical controls that isolate the incremental effect of a validator+repair floor on the same generated candidate. Moreover, published comparisons rarely quantify how sampling semantics (shared candidate versus independent draws) and repair budgets affect the number of informative discordant pairs available to paired binary tests. Surveys and architectural reviews therefore call for reproducible evaluation harnesses that combine deterministic validators with controlled sampling semantics and explicit inferential plans [10.6028/nist.ai.100-2e2023].

Our study addresses a tightly scoped portion of this gap by executing a preregistered paired experiment that (a) fixes shared_initial_candidate semantics to measure remediation capacity on a single generated plan, (b) uses a frozen deterministic validator to define a binary success criterion tied to intent constraints, and (c) limits repair budget to a single automated iteration. This design directly tests the operational claim that a minimal validator+repair floor reduces first-pass misconfiguration, and it surfaces the inferential consequences of high baseline pass rates and shared-candidate pairing for confirmatory McNemar-style testing.

III. METHODOLOGY

Preregistration and hypotheses: The experiment was preregistered (cand-2026-001-A-prereg-v1) prior to observing outcomes. Confirmatory hypotheses were: H1 — the guarded pipeline reduces first-pass misconfiguration rate by $\geq 50\%$ (relative) versus single-shot generation; H2 — one or fewer automated repair iterations recover $\geq 75\%$ of initially invalid configurations. The preregistration explicitly used a planning assumption of planned baseline pass rate = 0.40 for power calculations; this numeric assumption is central to the interpretation of the confirmatory outcome and is documented in the preregistration record (see Disclosure).

Design and sampling semantics: We adopted a paired within-task design with 300 deterministically generated seeds from a frozen task generator. For each seed the same sampled LLM candidate was reused in both arms (shared_initial_candidate semantics): baseline (single-shot generation) and guarded (identical candidate validated

and—if invalid—subjected to up to one automated repair attempt). Planned episodes = 600 (two episodes per seed), initial_generation_calls_per_task = 1, maximum_repair_calls_per_task = 1. The shared-candidate pairing was preregistered to isolate the incremental effect of validation and repair on a fixed generated plan; this choice increases within-pair correlation and reduces the number of informative discordant pairs available to McNemar inference compared with an independent-draws design.

Model, validator, and scoring: The declared model in the execution contract was gpt-5-mini. The frozen deterministic validator/repair adapter (validator id deterministic_netops_validator_v5) implements rule and dependency checks, canonicalizes configurations to normalized ASTs, and returns binary pass/fail scoring plus violation codes used to trigger and steer the preregistered automated repair adapter. The scoring rule is full intent-constraint validation: outputs are scored as success only when required command sequences and workflow constraints are satisfied by the validator’s canonicalization and rule checks. The preregistered missingness plan allowed one retry for provider failures; primary analysis used the ‘complete_pair_primary’ treatment which drops pairs only if both calls fail. All validator decisions, normalized configurations, and repair traces were archived in the execution manifest and analysis bundle.

Power, estimand, and analysis plan: The primary estimand was the paired_success_rate_difference_guarded_minus_baseline. The preregistered primary test was the exact McNemar two-sided test ($\alpha=0.05$) on paired pass/fail outcomes; we also preregistered paired bootstrap percentile 95% CIs (10,000 resamples) for the absolute paired difference. The preregistration included a planning assumption baseline pass rate = 0.40 which informed the sample size and the $\geq 50\%$ relative reduction target; this assumption is reported here to make the planning calculus transparent. Secondary analyses included exact binomial intervals for repair success, stratified checks, and sensitivity analyses treating failed model calls as failures; all analysis code and per-task artifacts are archived.

IV. RESULTS

Execution accounting and artifact provenance (archived): The preregistered run completed all planned episodes: planned episodes = 600, completed episodes = 600, complete analyzed pairs = 300. Model calls consumed = 301 total (300 shared initial generation calls + 1 repair invocation). The archived deterministic reconciliation/provider audit records show no provider or infrastructure failures and no deterministic exclusions for contamination under the preregistered checks. These execution counts and stage summaries are recorded in the archived execution manifest and analysis bundle and are the direct source for all numeric values reported below.

Primary per-condition outcomes (archived counts): Baseline (single-shot) success count = 299/300 (99.6667%); Guarded (validator + ≤ 1 repair) success count = 300/300 (100.0%).

The paired contingency table derived from the archived analysis artifact is: n_{11} (both success) = 299; n_{10} (guarded success, baseline fail) = 1; n_{01} (guarded fail, baseline success) = 0; n_{00} (both fail) = 0. Missingness summary in the archive reports zero failed or unscorable episodes in either arm under the preregistered missingness treatment.

Confirmatory inferential results (preregistered analyses executed on archived artifacts): The exact McNemar two-sided test on the observed contingency yields $p = 1.00$. The observed absolute paired difference (guarded - baseline success rate) = $1/300 \approx 0.0033333$. Paired bootstrap percentile 95% CI for the absolute paired difference = $[0.00, 0.01]$ (10,000 resamples, bootstrap seed = 1729), as computed in the archived analysis artifact. Repair outcomes: the single preregistered automated repair invocation succeeded on the sole initially invalid baseline output (repair success 1/1). The exact Clopper-Pearson 95% CI for a 1/1 repair success is $[0.025, 1.0]$, reflecting the large uncertainty from a single observed repair case.

Why the McNemar $p = 1.00$ and what the discordant count implies: Exact McNemar inference is driven entirely by the discordant pair counts (n_{10} and n_{01}). With only one discordant pair observed ($n_{10} + n_{01} = 1$), the two-sided exact test cannot reject the null of no difference; the minimal nonzero discordant total yields an exact two-sided p of 1.00 under the test implementation used in the preregistered analysis. Equivalently, the discordant total places a hard upper bound on the resolvable absolute paired difference equal to discordants / 300 = $1/300 \approx 0.00333$. Put practically, the observed near-ceiling baseline left essentially no statistical headroom to detect the preregistered $\geq 50\%$ relative reduction (which, given the preregistration's planning baseline = 0.40, mapped to an absolute change on the order of +0.30). Detecting an absolute +0.30 with $N=300$ would require on the order of tens of discordant pairs (the preregistration noted ~ 90 net discordant pairs aligned with the planned effect size under conservative assumptions); the archived discordant count of one is far below that requirement. These limits and numeric bounds are computed from the archived contingency artifact and bootstrap outputs.

Representative validator and diagnostic traces (artifact-grounded): For each episode the archived validator trace contains structured diagnostics used for scoring: `normalized_configuration`, `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `violation_codes`, `validation_before/after` states, and a `repair_applied` flag. In the solitary baseline failure the validator trace documents the missing required command sequence and the canonicalized configuration before and after the automatic repair; the post-repair canonicalized configuration satisfied the full intent constraints per the validator. These per-task traces (validator decision fields and repair flags) are the primary provenance for the contingency counts and are archived alongside the analysis artifacts.

Sensitivity and interpretive bounds (post-hoc, artifact-grounded): Because the observed discordant total is one, the maximal absolute improvement observable in this dataset equals $1/300$. The paired bootstrap CI $[0.00, 0.01]$

and the exact McNemar p together reflect that any larger effect size is incompatible with the observed discordant evidence. Operationally, this implies that confirmatory designs aiming to test large relative improvements must either (a) calibrate the task bank to produce a higher baseline failure prevalence or (b) adopt independent-draws sampling (increasing discordants at the cost of within-pair control) — choices that should be incorporated into power calculations. These precise numerical bounds are reproduced in the archived `paired_contingency_table.csv` and `paired_difference_figure.svg` contained in the analysis bundle.

Reproducibility and archived artifacts: All numerical counts, the bootstrap procedure (seed = 1729, resamples = 10,000), the exact-test implementation, and every per-task validator trace and repair prompt/response are included in the archived execution manifest and analysis bundle. The archived analysis artifacts (condition summary, paired contingency table, bootstrap outputs, and deterministic reconciliation/provider audit) reproduce the figures and numbers above when loaded with the provided analysis executor. Readers who wish to reproduce the exact inferential outputs can consult the execution manifest and the bundled analysis artifacts referenced in the Disclosure Statement.

Concise summary of the results section: The guarded pipeline corrected the single observed invalid baseline output (artifact-grounded repair success = 1/1) but, because the baseline pass rate was far higher than the preregistration planning assumption, the run produced only one discordant pair and therefore lacked statistical information to verify the preregistered $\geq 50\%$ relative reduction. The run nevertheless demonstrates end-to-end execution, archival, and artifact-grounded diagnostics for a generate→validate→repair NetOps pipeline and clarifies concrete sample-size and sampling-semantic constraints necessary for informative confirmatory inference in future studies.

V. DISCUSSION

Interpretation and operational implications: The guarded pipeline corrected the single observed invalid baseline output, demonstrating that a frozen deterministic validator plus a bounded automated repair iteration can remediate validator-detected violations in at least some instances. However, because the empirical baseline pass rate was far higher than the preregistered planning assumption, the run does not provide confirmatory evidence for the preregistered $\geq 50\%$ relative reduction. Practically, this implies that when baseline error rates are uncertain, confirmatory designs must focus on producing informative discordant pairs rather than simply increasing N under optimistic failure assumptions.

Sampling semantics and pairing tradeoffs: The preregistered `shared_initial_candidate` design intentionally reduced cross-arm variability to isolate validator/repair effects on a single generated plan. That choice is operationally motivated (real workflows often assess and remediate a single proposed plan) but reduces the number of statistically informative discordant pairs. Independent draws per arm would increase discordance

and therefore statistical information for McNemar tests but at the cost of within-pair heterogeneity. Future confirmatory studies should align pairing semantics with the targeted operational claim and incorporate their inferential consequences into power calculations.

Repair budget and conditional estimands: We treated repair effectiveness as H2 with a single allowed automated repair iteration. The observed repair success (1/1) is artifact-grounded but statistically imprecise. When repair effectiveness is a principal operational claim, we recommend treating repair success as a conditional estimand (conditional on initial failure) and powering the study to observe a sufficient number of initial failures to estimate that conditional probability with narrow confidence intervals.

Threats to validity and externality: Internal validity is affected by the preregistered `shared_candidate` semantics and the deterministic replacement rules in the preregistration. Construct validity is limited because the deterministic validator performs static canonicalization and rule checks rather than executing configurations in live devices; dynamic control-plane or packet-level consequences were not measured. External validity is limited by the synthetic task bank, constrained vendor template space, and a single model family (gpt-5-mini) and validator implementation. Conclusion validity is constrained by the paucity of informative discordant pairs (one), which leaves large preregistered effects unverifiable under the observed data. We document these threats and their artifact bases in the archived manifest.

Design recommendations for future confirmatory NetOps evaluations (artifact-grounded): (1) Calibrate task banks to target a modest initial-failure prevalence or a target discordant-pair count aligned with the planned effect size; concrete options include increasing task difficulty, adversarial augmentation, or stratified sampling by difficulty. (2) Explicitly preregister pairing semantics (shared candidate versus independent draws) and repair budgets and include their expected effect on discordant counts in power computations. (3) When operational claims require runtime semantic assurances, incorporate network emulation or simulation that exercises packet-level and control-plane consequences instead of relying solely on static validators. (4) Archive per-task validator traces and repair prompts systematically to enable post-hoc diagnostic analyses and reproducibility.

VI. LIMITATIONS

- Synthetic task bank and constrained vendor-template scope limit external validity to broad production environments.
- Single LLM (gpt-5-mini) and a single validator implementation restrict generality across model families and validator designs.
- The preregistered power plan assumed a baseline pass rate = 0.40; the observed baseline (299/300) contradicts that assumption and removed headroom to detect the preregistered $\geq 50\%$ relative reduction.

- Sampling semantics (`shared_initial_candidate`) reused the same initial candidate across paired arms, increasing within-pair correlation and reducing discordant pairs available to McNemar inference.
- Validation used deterministic configuration/constraint checks rather than full dynamic emulation of runtime semantic effects; actual runtime consequences of configurations were not executed and remain unmeasured.
- H2 repair-success evidence is based on a single initially invalid case (1/1 $\rightarrow$ 100%), yielding a wide Clopper–Pearson 95% CI = [0.025, 1.0]; larger counts of initial failures are required to estimate repair effectiveness precisely.

VII. CONCLUSION

We executed a preregistered paired experiment (N=300 pairs) comparing single-shot LLM generation (gpt-5-mini) to a guarded pipeline consisting of a frozen deterministic validator and a single automated repair iteration. Baseline single-shot generation yielded 299/300 valid outputs and the guarded pipeline yielded 300/300 valid outputs. The single automated repair corrected the lone invalid baseline output, but the observed near-ceiling baseline contradicted the preregistered planning assumption (baseline pass rate = 0.40) and removed statistical headroom to detect the preregistered $\geq 50\%$ relative reduction. Future confirmatory NetOps evaluations should (a) calibrate task difficulty to produce sufficient initial failures or target discordant counts, (b) preregister pairing semantics and repair budgets explicitly, and (c) include emulation to measure runtime semantic effects when operational deployment claims are intended. All raw outputs, validator traces, repair traces, the execution manifest, and analysis artifacts are archived and enumerated in the Disclosure Statement to permit reproduction of the reported counts and intervals.

DISCLOSURE STATEMENT

This study was executed autonomously after the autonomy lock under the archived master prompt (provenance/master_prompt.txt; SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). Human role: the Research Director selected the topic family (Generative AI and LLMs for NetOps) and approved the master prompt prior to the autonomy lock; no manual human editing of manuscript technical content, figures, tables, or references occurred after the lock. Preregistration: cand-2026-001-A-prereg-v1 (the preregistration record was created before observing final outcomes; preregistration SHA-256 is recorded in the execution manifest). Declared model and tooling: gpt-5-mini (declared_model_version), adapter family hosted_netops_gvr_v1, frozen deterministic validator/repair adapter deterministic_netops_validator_v5. Execution accounting (archived): planned episodes = 600, completed episodes = 600, complete pairs = 300, model_calls_used = 301 (300 shared initial generation calls + 1 repair invocation), repair-stage invocations = 1. The execution manifest and analysis artifact bundle

enumerate all raw model responses, per-task validator traces (normalized configurations, `parsed_command_count`, `required_command_count`, `violation_codes`, `repair_applied` flags), repair prompts/responses, execution logs, and analysis artifacts. The archived execution manifest and analysis bundle are the authoritative source for the numeric counts reported in this manuscript. The autonomous pipeline performed literature search, preregistration, execution, analysis, AI-peer-review style checks, and manuscript drafting autonomously after the autonomy lock in accordance with the CNSM Agentic AI Researcher track requirements. The preregistered planning assumption used in power calculations (planned baseline pass rate = 0.40) is explicitly reported here because it materially affects interpretation of the confirmatory result.

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